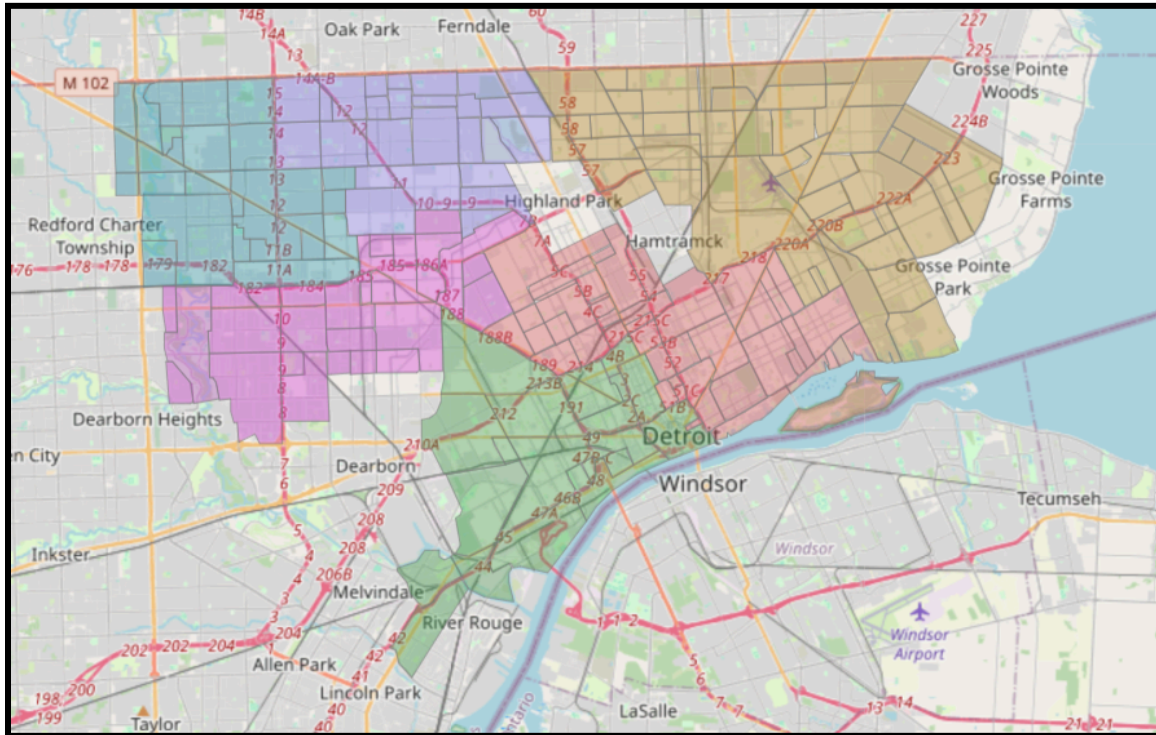


A GeoSpatial Analysis of Health Access, Service Requests, and Surveillance in Detroit

Evelyn Bartley and Megan Allen



Folium map of the 205 City of Detroit neighborhood boundaries color-coded by council district.

Motivation

Our project examines relationships among residential location, surveillance, health access, and non-emergency service requests in Detroit, with the goal of identifying neighborhoods and council districts for targeted, meaningful quality-of-life improvements. We both believe in the importance of studying socioeconomic factors to uncover and address existing disparities. Our shared interest in the Green Light Project discussed in the Data Ethics course led to a deep dive into Detroit's Open Data Portal [1], where we gathered data on Green Light surveillance locations [2], neighborhood boundaries [3], Medically Underserved Areas/Populations [4], and service request records from the Improve Detroit App [5]. Until this point, geospatial analysis of Green Light surveillance, Medically Underserved Areas/Populations, and Improve Detroit requests has not been conducted. Through the integration of these datasets, our analysis provides multifaceted insight for city planning efforts in Detroit.

Similar Works

In gathering our sources for this project, we identified several studies and articles that informed our approach.

- The University of Michigan Deep Blue Data project titled “Neighborhood Effects: Community Characteristics and Health in Metropolitan Detroit” aimed to “develop and validate new metrics of neighborhood factors that can help explain health outcomes” and resulted in a database of food, health, housing, education, living resources, and social data [6]. Their exploration provided an intellectual template for our examination of community-level impacts.
- The ArcGIS StoryMaps webpage titled “Turning The Corner: Local Analytical Findings from Detroit” [7] was foundational in our approach to this project. It is a multi-city initiative led by the Urban Institute’s National Neighborhood Indicator Partnership, and presents its findings in a combination of narrative and computer-generated maps.
- Detroit’s Master Plan of Policies (Plan Detroit) is a “roadmap for Detroit’s future. It looks at where we are now, what we hope to be in the future, and what policies and actions are needed to achieve that shared future vision” [8]. There are numerous current projects for Detroit’s growth, and this is a major government-level initiative that groups neighborhoods together to “align strategic decision-making about when, where, and how investments, capital improvements, and department initiatives will be implemented with the Plan’s vision”. We hope that our work could be informative for a project like this.
- The January 12, 2026 article titled “Mayor Sheffield taps UM Poverty Solutions founder to be Detroit’s first Chief of Health, Human Services, and Poverty Solutions” is an announcement of a “sweeping reorganization plan to improve the delivery of critical supportive services to Detroiters most in need to help break the cycle of poverty and enhance their quality of life” [9]. The plan is to “bring national best-practices to the City’s approach to addressing household basic needs, lowering mental, physical and behavioral health barriers, and leading an administration-wide approach to addressing the root causes of poverty in Detroit” [9].
- The University of Michigan’s Center for Ethics, Society and Computing compiled a lesson called “Race, Policing, and Detroit’s Project Green Light” that was shared in the MADS Data Science Ethics class. The publicly-available version of this lesson [10] includes a video interview with Tawana Petty, former director of the Algorithmic Justice League, and addresses issues of wrongful arrests from flawed facial recognition algorithms, as well as the disparities that result to disadvantaged communities from a paid “priority” response system.

Data Sources

Current City of Detroit Neighborhoods

URL: <https://data.detroitmi.gov/datasets/detroitmi::current-city-of-detroit-neighborhoods/about>

The Current City of Detroit Neighborhoods dataset was accessed through the City of Detroit Open Data Portal and is described as “Current (non-historic) neighborhood boundaries as compiled by Department of Neighborhoods staff in concert with community groups” [3]. There were several download options, including CSV and Shapefile. We downloaded it as a GeoJSON file to use the neighborhood boundary polygon geometries it contained (other formats, including CSV, didn’t include geometry data). Important variables in the file include the council_district, nhood_number, nhood_name, and geometry. There are 205 records, each corresponding to a Detroit neighborhood, compiled by the Department of Neighborhoods and community groups. There is no time element to the data.

Project Green Light Locations

URL: <https://data.detroitmi.gov/datasets/detroitmi::project-green-light-locations/about>

According to the City of Detroit website, Project Green Light is the “first public-private-community partnership of its kind, blending a mix of real-time crime-fighting and community policing aimed at improving neighborhood safety, promoting the revitalization and growth of local businesses, and strengthening DPD’s efforts to deter, identify, and solve crime” [11]. The Project Green Light Locations dataset was accessed through the City of Detroit Open Data Portal [2]. It identifies and locates local businesses and organizations that have “installed real-time camera connections with Detroit Police Department headquarters and visibly communicate[d] their participation to the public through project-specific green lights and signage at each participating location” [11].

There were several download options, including CSV and Shapefile. We downloaded the dataset as a GeoJSON file to use its point location geometries (other formats, including CSV, didn’t include the geometry data). Important variables in the file include OBJECTID, business_name, business_type, precinct, greenlight_live_date, and geometry. There are 1,085 records, each corresponding to a Green Light surveillance system location. The greenlight_live_date variable denotes the date on which the Green Light location went live. The live dates span from January 1st, 2016, to the present.

Medically Underserved Areas/Populations

URL: <https://data.detroitmi.gov/datasets/detroitmi::medically-underserved-areas-population/about>

The Medically Underserved Areas Population dataset is available through the City of Detroit Open Data Portal; it records areas and populations within the City of Detroit designated as Medically Underserved Areas/Populations (MUA/Ps) [4]. The MUA/P designation derives from the Index of Medical Underservice (IMU) score, which is calculated by the U.S. Health Resources & Services Administration. An area’s Index of Medical Underservice score is calculated from four sources: the population-to-provider ratio, percent of the population below the federal poverty level, percent of the population over age 65, and infant mortality rate [13]. The IMU Index ranges from 0 to 100, with lower scores indicating greater medical underservice. Since this dataset consists only of areas/populations within the City of Detroit with an MUA/P designation, there are no records for areas and populations with an IMU score exceeding the MUA/P designation threshold of 62 [4].

There were several download options, including CSV and Shapefile. We downloaded it as a CSV file to retain the in-depth information that the GeoJSON file lacked. Important variables in the file include

medical_underservice_ranking (or Index of Medical Underservice), the date when the area was designated as medically underserved, the GeoID and census tract information, as well as the latitude and longitude of the centroids.

Improve Detroit Issues

URL: <https://data.detroitmi.gov/datasets/detroitmi::improve-detroit-issues/about>

The Improve Detroit Issues dataset was accessed through the City of Detroit Open Data Portal. The dataset consists of issues from the Improve Detroit application, an application that Detroiters can use to submit “non-emergency service requests to the City and to check the status of non-emergency requests such as potholes, graffiti and damaged street signs” [5]. The description on the Apple App Store says, “For reporting running water, potholes, damaged street signs, and other issues, Improve Detroit makes reporting a neighborhood problem to City Hall easier than ever. The app uses your location and provides a menu of common quality-of-life issues to choose from in your service request. Uploading a photo with your report also helps the City staff quickly identify the problem. You may comment on your submissions, and you will be notified when the status of the issue is updated” [14].

Due to the size of this 680,000+ record dataset, we imported it through the API Explorer embedded in the Detroit Open Data Portal [15], chunking our request by Issue ID. Important variables in this file include the neighborhood and council district, as those are our primary categories to group and aggregate by, as well as the Issue ID as an index of the requests, the type of request, the priority code, and the latitude and longitude columns that became our geometry column.

Data Manipulation Methods

Workflow

The code for this project is in a Python Notebook (.ipynb file) that contains a balance of Markdown narrative and code. The notebook is organized into four main sections: Notebook Initialization, Load Data, Data Processing and Manipulation, and Data Analysis. The Notebook Initialization section consists of the notebook dependencies summary, code to import and load the required packages, and code to import our functions module from our public GitHub repository. The Load Data section includes links to each dataset’s source website, code that loads data files in from GitHub and saves as dataframes, and code that requests, retrieves, and saves data from an API.

Data Processing

The Current City of Detroit Neighborhoods dataset was processed and manipulated into a GeoDataFrame (named `current_neighborhoods_gdf`) with a record for each unique neighborhood in the City of Detroit. After manipulation, `current_neighborhoods_gdf` had five columns: `council_district` (council district of the neighborhood, data type: str), `nhood_number` (renamed to `neighborhood_number`; the unique numeric ID of the neighborhood; data type: str), `nhood_name` (renamed to `neighborhood_name`; the unique name of the neighborhood; data type: str), and `geometry` (the GeoJSON geometry Polygon data; data type;

geometry). We also added a custom column titled `area_sq_miles` (data type: `int`) that stores the area (in square miles) of each neighborhood.

The Project Green Light Locations dataset was named `green_light_gdf` and was processed and manipulated into a `GeoDataFrame` with `Location_id` (the unique numeric ID of the location; data type: `str`), `business_name` (the business name where the camera is located; data type: `str`), `business_type` (the type of business; data type: `str`), `precinct` (the police precinct the camera is located in; data type: `str`), `greenlight_live_date` (the date the camera went live; data type: `datetime(64)`), and `geometry` (the GeoJSON geometry Point data; data type: `geometry`).

From a copy of `green_light_gdf`, we created another `GeoDataFrame` that contained the same information as `green_light_gdf`, with an additional two columns. Through a spatial join (using `.sjoin`), we added `neighborhood_name` (the neighborhood that a Green Light Point resides in; data type: `str`), and `council_district` (the council district that a Green Light Point resides in; data type: `str`), and named the new `GeoDataFrame` `green_light_points_with_neighborhoods`.

	Location_id	business_name	business_type	precinct	greenlight_live_date	geometry	neighborhood_name	council_district
0	1	Sunoco	Retail	2	2019-12-04	POINT (-83.17978 42.40904)	Bethune Community	2
1	2	New Center Pavilion	Residential	3	2019-12-03	POINT (-83.07918 42.37103)	New Center Commons	5
2	3	New Flamingo Liquor	Retail	10	2019-11-29	POINT (-83.12104 42.36905)	Nardin Park	7
3	4	Citgo	Retail	11	2019-11-29	POINT (-83.10204 42.43984)	Nolan	3
4	5	Mack Ashland Apartments	Residential	5	2019-11-27	POINT (-82.94933 42.38731)	Fox Creek	4

The first five rows of `green_light_points_with_neighborhoods`.

The Medically Underserved Areas/Populations (MUAP) dataset was loaded into our environment as a `pandas DataFrame`. Since the `DataFrame` had latitude and longitude coordinates (named `intptlon_10` and `intptlat_10`), we defined a function (`df_to_geodataframe`) that takes a `DataFrame` with coordinate columns, creates a 'geometry' column from the coordinates and converts it to a `GeoDataFrame` in the specified coordinate reference system. We used the World Geodetic System (`crs = "EPSG:4326"`), a global coordinate reference system. Once we had the `GeoDataFrame`, `medically_underserved_gdf`, we removed unnecessary columns and changed the data type of columns we were keeping. This left us with `ObjectId` (the unique ID for the area/population; data type: `int64`), `muap_index` (renamed to `med_underserv_ranking`; the IMU Index value of the area/population; data type: `float64`), and `geometry` (the GeoJSON Point of the area/population; data type: `geometry`).

From a copy of `medically_underserved_gdf`, we created another `GeoDataFrame` that contained the same three columns in the original, and an additional three columns. Through a spatial join (using `.sjoin`), we added `neighborhood_name` (the neighborhood that a MUAP Point resides in; data type: `str`), `council_district` (the council district that a MUAP Point resides in; data type: `str`), and `area_sq_miles` (the area in square miles of the neighborhood an MUAP Point resides in; data type: `float64`). The new `GeoDataFrame` was named `med_undserv_with_neighborhoods_gdf`.

The Improve Detroit Issues dataset was retrieved from the API request and read into the notebook as a pandas DataFrame. This DataFrame had latitude and longitude coordinates (named latitude and longitude), so we used our function `df_to_geodataframe` to convert it to a GeoDataFrame (`improve_detroit_gdf`). Next, we removed unnecessary columns and changed the data type of columns we were keeping. This left us with `ObjectId` (the unique ID of the request; data type: `int64`), `request_type` (the type of request; data type: `str`), `neighborhood` (the neighborhood the request was associated with; data type: `str`), `council_district` (the council district the request was associated with; data type: `str`), and `geometry` (the GeoJSON Point of the request location; data type: `geometry`).

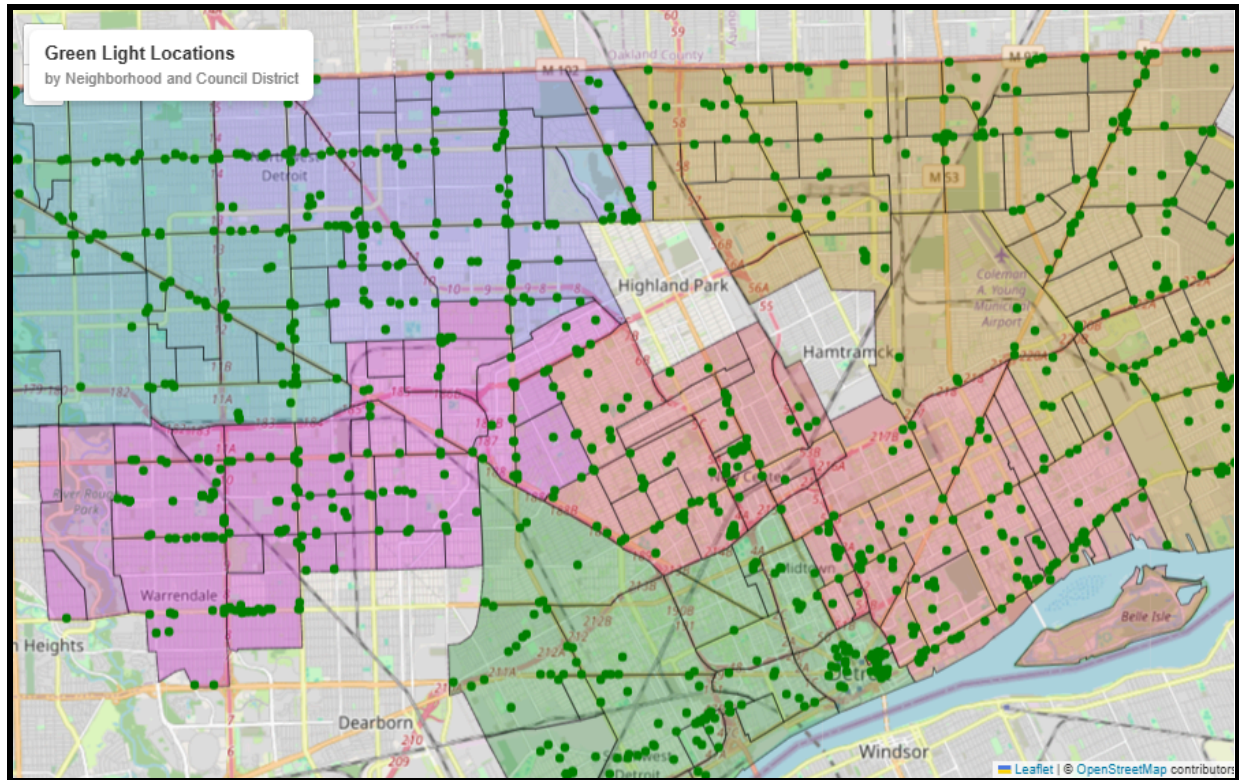
Challenges

The first challenge was determining which neighborhood boundaries to use and why. Many of our datasets contain names and polygon boundary information for the current City of Detroit neighborhoods, but we found that there are multiple types of neighborhood boundaries, each for different purposes. We spent much time experimenting with different neighborhood boundaries (larger-granularity Master Plan neighborhoods to align with current city-level projects, smaller-granularity Census Tract boundaries that ultimately did not align across datasets). Ultimately, the most effective approach was to use the 205 current city of Detroit neighborhood boundaries and aggregate them by council district to provide our findings to a potential local government audience.

The Improve Detroit dataset presented an overwhelming hurdle: 94% of the records lacked geographic location data. The vast majority of people reporting service requests entered their requests using a physical address or general neighborhood labels; therefore, processing this data involved grouping by neighborhood, counting requests by neighborhood, and then aggregating by council district. Even the 6% of records with geographic location data (~41,000 records) proved too large for our Folium maps. For this dataset, instead of a map showing the locations of all 41,000 service requests, we created a choropleth map that grouped the requests by neighborhood and highlighted the neighborhood with the highest density of service requests per mile.

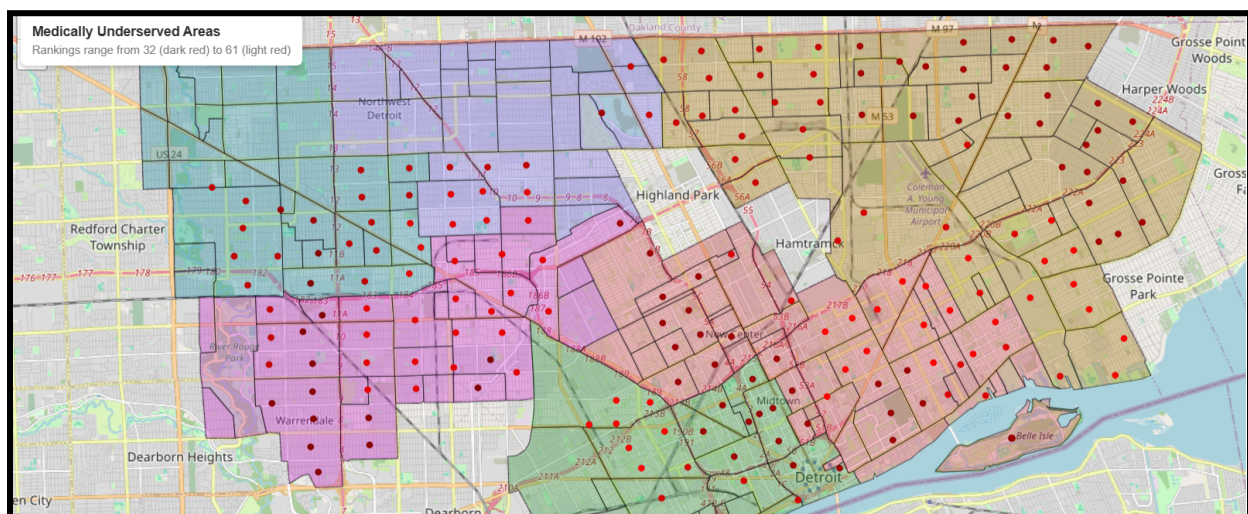
Analysis

Our analysis was guided by the nature of our data. Each dataset included a certain level of location data, so we decided to integrate our data through geospatial location. We wrote a function that creates a Folium map that displays (on toggle) the Green Light Locations, Medically Underserved Areas/Populations points and Improve Detroit service requests layered on the map of Current Neighborhoods and council districts. It includes an interactive hover menu that displays (on hover) the neighborhood name, council district, and relevant information for map points, such as the business name at the surveillance camera location or the area/population's IMU score.



Green Light Locations overlaid on top of the Current Neighborhoods, color-coded by council district.

For our analysis, it was important to determine whether a neighborhood was “residential”, “commercial”, or “mainly parkland”. It is reasonable for a small commercial neighborhood like Greektown to have 15 Green Light surveillance cameras in a 0.05-square-mile area. In contrast, having 76 surveillance cameras per square mile, as in the New Center Commons residential area, might feel oppressive and potentially threatening, especially in the context of today’s immigration raids.



Medically Underserved Areas/Populations points, with darkness representing index of medical underservice, overlaid on top of the current neighborhoods, which are color-coded by council district.

The Medically Underserved Areas dataset was interesting because not every neighborhood in Detroit is medically underserved. Districts with a medical underservice ranking above 62 (areas that are not “underserved”) were not included in the dataset. To generate meaningful insights about the council districts, our “calculate_district_imu” function required a “fallback” score to fill in for neighborhoods without medically underserved points. With all the underserved neighborhoods ranking between 32 and 62 on the Index of Medical Underservice, the fallback number we chose for neighborhoods outside of that range was 70— enough to pull that council district away from the lowest rankings without masking the lower scores that might be there.

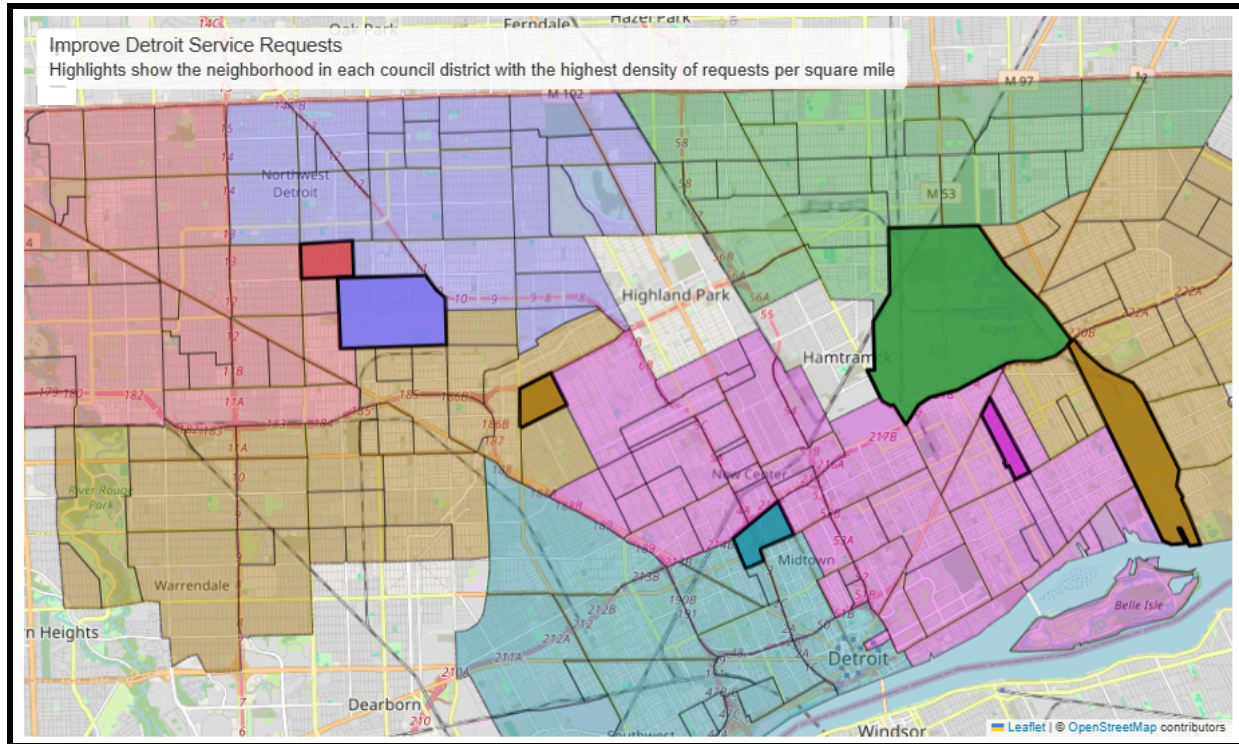
The Medically Underserved Areas dataset also required a “non-residential neighborhoods” list. Belle Isle is a large park that ranked at the top of the medically underserved data (the most medically underserved neighborhood), yet it has no full-time residents. Eliminating parkland and commercial districts significantly affected our insights into this dataset.

We computed point density per neighborhood (points per square mile) to quantify spatial concentration patterns and evaluate where surveillance cameras are most concentrated, which neighborhoods are most medically underserved, and which areas generate the highest volume of service requests. The medical underservice dataset complicated these plans because the areas are represented by centroid points in a grid-like layout across the city; thus, point density was less relevant than the medical underservice ranking of the points, which corresponded to census tracts rather than neighborhoods.

We aggregated information on points in each neighborhood using a “council_district_stats” function that displayed, for each council district, the neighborhood with the minimum and maximum density and the number of points, and computed the average points per square mile to compare the seven council districts.

Finally, we created a map of neighborhoods that highlighted the council districts with the highest density (points per square mile) for each point type. This allowed the user to visually identify patterns between neighborhoods and council districts, thereby increasing awareness of where problems tend to concentrate and enabling decision-makers to channel efforts, human resources, and funding to remedy these areas.

The Improve Detroit Data introduced a few types of bias. Sampling bias is evident; individuals who submit requests through the Improve Detroit app don’t necessarily represent the general population. There is non-response bias; the Improve Detroit App is publicly available, though not everyone owns or has access to the internet and/or a device. The overwhelming majority of records that lack location data may be due to data-collection errors, which would constitute measurement bias.



Folium map of highlighted neighborhoods.

We anticipated that we would see patterns in council districts or neighborhoods repeatedly coming up as over-surveilled, medically underserved, and having many city service projects. The reality of our findings, however, was that there was no clear pattern. Council District 7 had the highest number of Green Light locations, but Council District 5 had the highest density per square mile. Council District 6 had the lowest area-weighted Index of Medical Underservice (at 47), but there were neighborhoods in the low to mid-30s in most council districts. Russell Woods neighborhood in Council District 7 had a significantly higher maximum density of Improve Detroit service requests at 10,203 requests per square mile, but all council districts also had neighborhoods with zero service requests.

Statement of Work

Evelyn Bartley

- Equal contribution to project formation and the search for relevant data
- Loaded data from the GitHub repository into the notebook
- Wrote and optimized the API request to load the Improve Detroit Issues data
- Compiled and implemented the functions module
- Led data processing and manipulation
- Created distribution visualizations in EDA
- Integrated markdown chunk narratives and technical code comments throughout the code notebook
- Guided code notebook iterations through Colab, DeepNote, and Jupyter
- Polished code notebook and functions module to ensure PEP8 compliance
- Primary author of Motivation, Data Sources and Data Manipulation Methods sections
- Wrote the README.txt and compiled the zip file

Megan Allen

- Equal contribution to project formation and the search for relevant data
- Researched government-level sources to learn about census tract data, the index of medical underservice, and Master Plan city improvement projects
- Compiled tables of column data to understand government identifiers such as “GeoID” and census tract codes
- Defined the majority of our functions
- Implemented the functions in the notebook
- Created summary dataframes for each dataset
- Primary author of the Analysis section
- Led the direction and implementation of the analysis
- Wrote the functions to display GeoDataFrame geometry data with Folium maps

Collaboration Summary:

Our team was effective in communication and collaboration. We attended Google Meet sessions at least twice per week, where we shared ideas and asked one another questions about the processes we were developing. Our contributions were often layered and overlapping, with one person taking the lead and the other picking up the baton to carry the next steps forward. In this way, we both contributed to all parts of the report and code, continually refining our shared processes and outputs.

If we were to collaborate again, we would select a single coding environment that meets our needs and continue refining our processing and EDA pipeline to maximize efficiency with the datasets at hand. Much more could be done with this wealth of data if we had more time. Overall, we are very proud of our shared contributions to this project.

References

- [1] City of Detroit Open Data Portal. “Welcome to Detroit’s Open Data Portal”. <https://data.detroitmi.gov/> (accessed February 1, 2026).
- [2] City of Detroit Open Data Portal. “Project Green Light Locations”. <https://data.detroitmi.gov/datasets/detroitmi::project-green-light-locations/about> (accessed February 1, 2026).
- [3] City of Detroit Open Data Portal. “Current City of Detroit Neighborhoods”. City of Detroit Open Data Portal. <https://data.detroitmi.gov/datasets/detroitmi::current-city-of-detroit-neighborhoods/about> (accessed February 1, 2026).
- [4] City of Detroit Open Data Portal. “Medically Underserved Areas Population”. City of Detroit Open Data Portal. <https://data.detroitmi.gov/datasets/detroitmi::medically-underserved-areas-population/about> (accessed February 1, 2026).
- [5] City of Detroit Open Data Portal. “Improve Detroit Issues”. City of Detroit Open Data Portal. <https://data.detroitmi.gov/datasets/detroitmi::improve-detroit-issues/about> (accessed February 1, 2026).
- [6] T. Veinot, V. Berrocal, P. Clarke, R. Goodspeed, D. Romero, and V. V. Vydiswaran, “Neighborhood Effects: Community Characteristics and Health in Metropolitan Detroit,” *Deep Blue Data*, University of Michigan. <https://deepblue.lib.umich.edu/data/collections/pk02c9792> (accessed February 1, 2026).
- [7] “Turning The Corner,” *ArcGIS StoryMaps*. <https://storymaps.arcgis.com/stories/9ef0f0dcb74e4cbc8a4bc09c9bdba9f4> (accessed February 1, 2026).
- [8] “Plan Detroit — Learn More,” *Plan Detroit*. [Online]. Available: <https://plan-detroit.webflow.io/> (accessed February 1, 2026).
- [9] City of Detroit, “Mayor Sheffield taps UM Poverty Solutions founder to be Detroit’s first Chief of Health, Human Services and Poverty Solutions,” *City of Detroit*. <https://detroitmi.gov/news/mayor-sheffield-taps-um-poverty-solutions-founder-be-detroits-first-chief-health-human-services-and> (accessed February 1, 2026).
- [10] “Race, Policing, and Detroit’s Project Green Light,” *Emergency ESC*, University of Michigan. <https://esc.umich.edu/project-green-light/> (accessed February 1, 2026).
- [11] City of Detroit. “Project Green Light Detroit”. City of Detroit. <https://detroitmi.gov/departments/police-department/project-green-light-detroit> (accessed February 14, 2026).
- [12] City of Detroit, *Detroit Health Department Locations – image*, City of Detroit. <https://detroitmi.gov/departments/detroit-health-department/detroit-health-department-locations> (accessed February 1, 2026).
- [13] U.S. Health Resources & Services Administration (HRSA). "Shortage Designation Scoring" <https://bhwh.hrsa.gov/workforce-shortage-areas/shortage-designation/scoring> (accessed February 14, 2026).
- [14] Apple Inc. n.d. “Improve Detroit.” App Store. <https://apps.apple.com/us/app/improve-detroit/id879463277> (accessed February 1, 2026).
- [15] Esri, “Improve Detroit Feature Service – Query API (JSON).” https://services2.arcgis.com/qvkbeam7Wirps6zC/arcgis/rest/services/improve_detroit/FeatureServer/0/query?where=1%3D1&outFields=*&outSR=4326&f=json (accessed February 1, 2026).
- [16] City of Detroit Department of Public Works, “We strive for cleanliness, safety, and efficiency,” *City of Detroit*. <https://detroitmi.gov/departments/department-public-works> (accessed February 1, 2026).

[17] R. J. King, “Rendering of Greektown streetscape,” in *Detroit’s Greektown Lands \$20M Redevelopment Grant to Boost Business, Improve Safety*, DBusiness Magazine, June 30, 2023. <https://www.dbusiness.com/daily-news/detroits-greektown-lands-20m-redevelopment-grant-to-boost-business-improve-safety/> (accessed February 1, 2026).

Image Appendix



A Green Light surveillance camera in Detroit. Source: [10]



Image from the Tawana Petty interview in the UMich Center for Ethics, Society and Computing lesson. Source: [10].

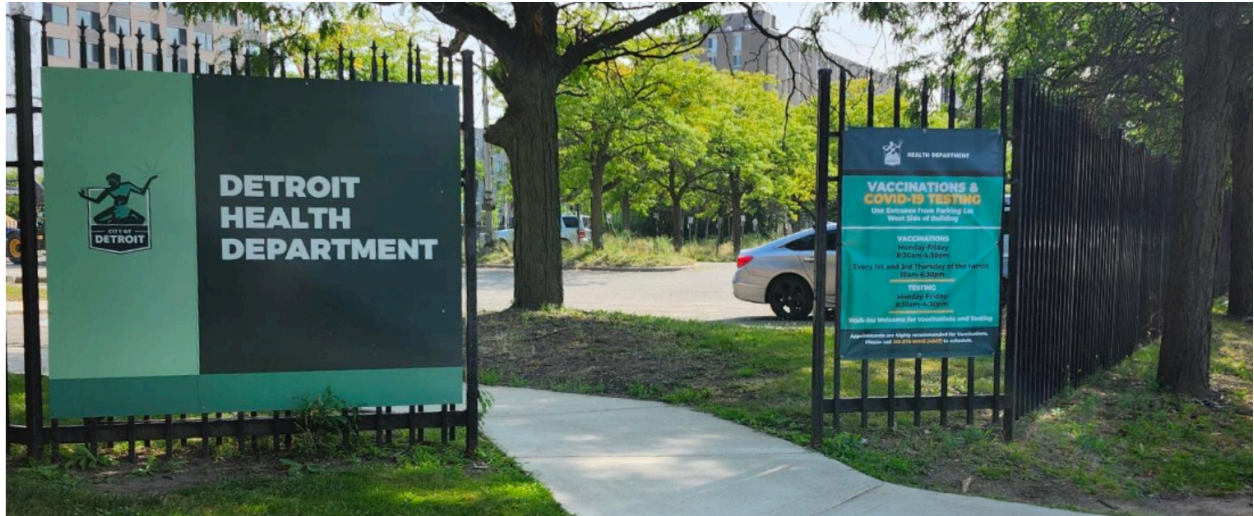


Image of Detroit Health Department sign. Source: [12]

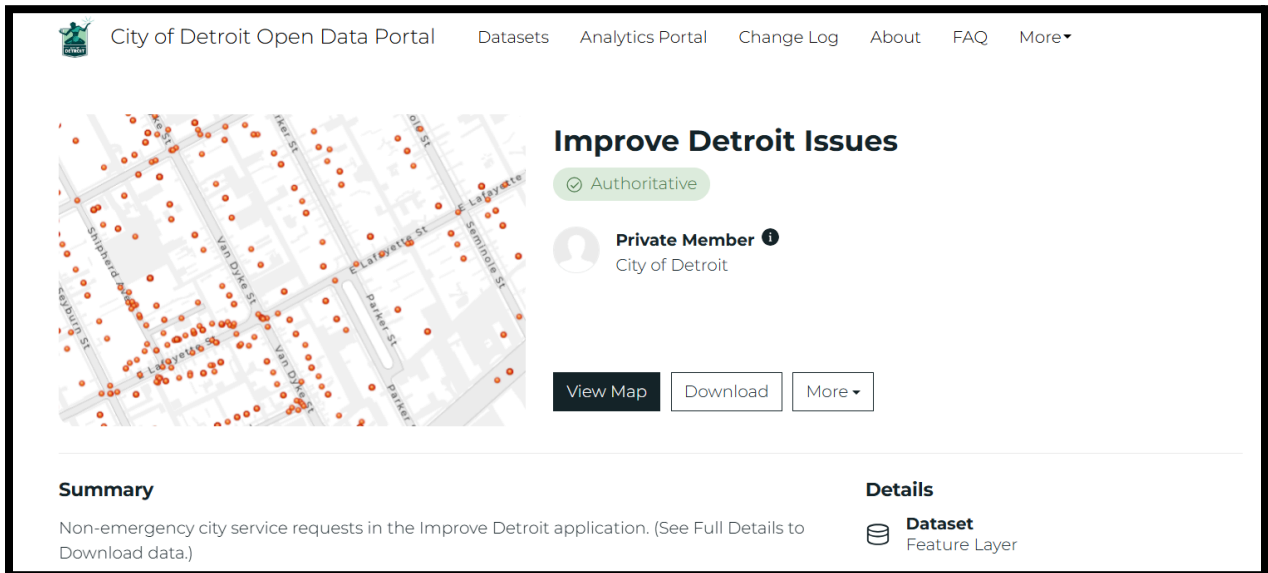


Image from the City of Detroit Open Data Portal. Source: [5].



We strive for cleanliness, safety, and efficiency.

Image from City of Detroit Department of Public Works. Source: [16]



Rendering of a new \$20M redevelopment project in Detroit's Greektown. Source: [17].